

Android Based Invisible Broken-Wire-Detection Robot



JAYANTH MURALA

Electrical conductors for distribution are carried on overhead pole lines or buried underground. It is easy to identify broken electrical wires on overhead lines but not so in underground wires or household wirings concealed in the walls or breakage inside their insulation cover. Here is a robot that can detect invisible broken electrical wires in 1-core, 2-core and 3-core cables without physically disturbing these. Author's prototype is shown in Fig. 1.

The robot is controlled using Bluetooth with an Android app. It can travel on ground, tracing an AC line cable to detect a broken conductor wire or breakage point.

Circuit and working

Block diagram of the invisible broken wire-detection robot is shown in Fig. 2. It consists of an Android mobile device and robot sections, which further consist of controller and detector units.

Fig. 3 shows the controller section with Bluetooth and Arduino interfacing circuitry. The project includes Arduino UNO board, JY MCU BT Bluetooth module, L293D motor driver IC, DC motors M1 and M2 and a few other components. The 9V battery used is to power Arduino board through Vin terminal. Regulated 5V supply is provided by Arduino board itself for rest of the circuit.

Bluetooth module. JY MCU BT Bluetooth module is used in this project (Fig. 4). It can be connected to any device via built-in UART interface, to communicate with other Bluetooth-enabled devices such as mobile phones, handheld computers

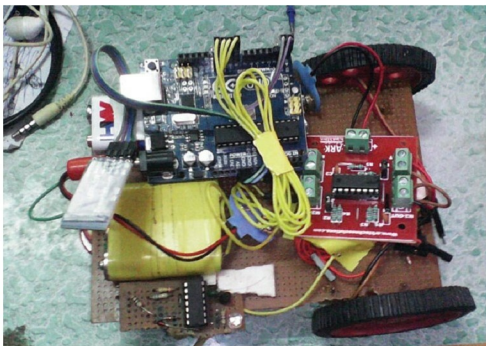


Fig. 1: Author's prototype

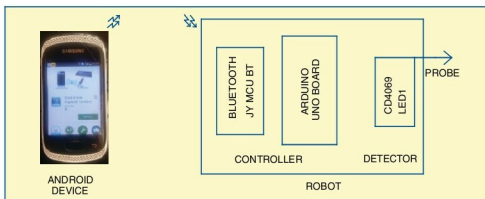


Fig. 2: Block diagram of the Android based invisible broken wire-detection robot

and laptops.

Arduino UNO board. Arduino is an open source electronics prototyping platform based on flexible, easy-to-use hardware and software. It is intended for artists, designers, hobbyists and anyone who may be interested in creating interactive objects or environments.

Arduino UNO is a board based on ATmega328 microcontroller (MCU). It has 14 digital input/output pins, six analogue inputs, a USB connection

for programming the onboard MCU, power jack, an ICSP header and a reset button. It is operated with a 16MHz crystal oscillator and contains everything needed to support the MCU. The board is very easy to use as the user simply needs to connect it to a computer using a USB cable or power it with an AC-to-DC adaptor or battery to get started. The MCU on the board is programmed using Arduino programming language using Arduino development environment.